MANISH D GANVIR

Python Network Automation Engineer

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https://www.linkedin.com/in/manishganvir/

https://github.com/MANISH-GANVIR/vpn-automation-framework

Language Known: English

Bengaluru, KA 560043

Network Automation and Security professional with **over 6 years of experience**, specializing in **Python-based automation across application and network layers**. Possess strong hands-on expertise in **building automation solutions for VPN operations, firewall policy management, and network troubleshooting** using **Python and APIs**. Experienced in **integrating backend automation services** with **web interfaces** and managing **enterprise and cloud-connected networks**. Seeking a role to leverage **full-stack automation skills** to deliver secure, scalable, and reliable network platforms.

Profile Summary

- Experienced in designing, implementing, and automating VPN solutions, including **IPsec Site-to-Site, SSL AnyConnect & GlobalProtect**, across multi-vendor enterprise environments.
- Skilled in **Python-based network automation**, building frameworks for VPN lifecycle management, firewall policy automation, and log collection using **Netmiko, Paramiko, NAPALM, PAN-DEVICE, and PAN-OS APIs**.
- Proficient in configuring and managing enterprise firewalls such as **Palo Alto, Cisco ASA, Fortinet, and Sophos**, with a strong focus on security policy enforcement.
- Strong expertise in **routing and switching technologies**, including BGP, OSPF, VLAN, EtherChannel, and HSRP for LAN and WAN environments.
- Hands-on experience with **cloud & on-premises data center networks**, including data center migrations, high-availability configurations, firmware upgrades & performance monitoring.
- Adept at integrating network automation with **CI/CD pipelines, vulnerability scanning tools such as Rapid7**, and compliance validation processes to improve operational efficiency.
- Experienced in troubleshooting and monitoring network health, including **IPsec VPNs, firewall policies, NAT validation, SSL decryption, and URL filtering**.
- Proven track record of **reducing manual operational effort by approximately 60%** and improving network reliability through automation and standardized processes.
- Strong knowledge of **network security architecture, IDS/IPS**, and enterprise firewall policy enforcement to ensure secure traffic flow across multi-vendor environments.
- Excellent **team collaboration, project execution, and documentation skills**, with experience delivering solutions across enterprise, cloud, and data center environments.

Certifications & Professional Training

- **Certified Ethical Hacker (CEH V11) - EC-Council (ECC9746051832)**
- **Cisco Certified Network Professional (CCNP - R/S)**
- **Cisco Certified Network Associate Security (CCNA Security)**
- **Cisco Certified Network Associate (CCNA)**
- **Python Full-Stack Development**

Technical Skills

Automation & Scripting:

- **Python for Network Automation:** Netmiko, Paramiko, NAPALM, PAN-OS SDK, pan-device, pan-python
- **APIs and Integrations:** PAN-OS XML APIs, PAN-OS REST APIs, FastAPI, Panorama for monitoring and centralized policy deployment
- **Frontend Technologies:** HTML, CSS, JavaScript; web-based UI integrated with FastAPI backend for real-time VPN automation operations
- **Automation Projects:** VPN automation for operations such as reset, upgrade, build, and troubleshooting; firewall rule and object management automation; log collection and troubleshooting frameworks

Network Security & VPN:

- **VPN Technologies:** IPsec Site-to-Site, SSL VPN including AnyConnect and GlobalProtect, IKEv1, IKEv2, IKE Crypto Profiles, IKE Gateways, IPSec Crypto Profiles, Tunnel Interfaces, Security Policies
- **Firewall Platforms:** Palo Alto Networks PAN-OS, Cisco ASA, Fortinet, Sophos, Cyberoam
- **Security Concepts:** IDS/IPS, NAT, U-Turn NAT, SSL Decryption, URL Filtering

Networking & Infrastructure:

- **Routing and Switching:** BGP, OSPF, VLAN, HSRP, EtherChannel (LACP, PAgP)
- **Data Center and Cloud:** On-premises to Azure connectivity, cloud operations, firewall high availability, firmware upgrades

Ongoing Skills & Learning:

- Azure Networking (AZ-700) - Hands-on practice
- GitHub Actions - CI/CD workflow automation for Python projects
- Advanced FastAPI - Backend optimization and API security

Core Competencies

Network Automation & Process Standardization

VPN Design, Implementation & Troubleshooting

Automation of Palo Alto Networks IPSec VPN using pan-os-python SDK

Automation of Cisco ASA IPSec VPN Operations using Netmiko over SSH

Network Security & Threat Management

Firewall Policy Enforcement & Governance

Enterprise Network Operations & Monitoring

Data Center & Cloud Network Connectivity

Network Infrastructure Planning, Upgrades & Migration

Performance Optimization & Reliability Management

Incident Detection, Response & Vulnerability Management

Risk Assessment & Compliance Management

High Availability, Redundancy & Business Continuity Planning

Change & Configuration Management

Routing, Switching & VLAN/IP Addressing Management

Multi-vendor Network Coordination & Integration

Project Planning, Stakeholder Collaboration & Reporting

Education & Credentials

- **Bachelor of Engineering (B.E.) - Information Technology** | RGCER, Nagpur, India | 2016

  ***Final Year Project**: Explored Evolutionary Algorithms for Optimal Feature Selection in Remote Sensing Satellite Images*

Soft Skills

- **Strong Analytical & Problem-Solving Skills**
- **Excellent Documentation & Communication Skills**
- **Effective Team Collaboration & Stakeholder Coordination**
- **Ownership-Driven & Automation-Focused Mindset**

Current Project Details

VPN Automation Framework:

- Designed and developed a **Python-based multi-vendor VPN automation platform**, automating end-to-end VPN lifecycle operations including Build, Update, Reset, Troubleshooting, IKE/IPsec SA resets, and tunnel health checks.
- Implemented **CLI automation for Cisco ASA** using Netmiko and an **extensible architecture for Palo Alto Networks** using PAN-OS APIs, integrated with a **web-based UI** built with FastAPI, HTML, CSS, and JavaScript.
- Delivered **modular, reusable, and consistent automation workflows**, reducing manual VPN operations, enhancing troubleshooting accuracy, and improving overall network efficiency.

Work Experience

Network Automation Engineer | Epicor Software India Pvt. Ltd. — Bangalore, India | Oct 2025 - Present

- Working on an **Enterprise VPN Automation Platform** designed to streamline large-scale firewall and VPN operations across multi-vendor environments.
- Designed and developed a **Python-based VPN automation platform** with a **user-friendly login portal (frontend UI)** enabling users to select rule-based projects and execute automated firewall operations.
- Implemented a **backend automation framework in Python** to handle the full VPN lifecycle, including:
  - VPN **Build**, **Update**, **Reset**, and **Troubleshooting**
  - **Health checks**, **PSK/ACL updates**, **NAT checks**
  - **Security Association (SA) resets**, **IKE/IPsec tunnel analysis**
- Automated **multi-vendor firewall operations**, significantly reducing manual intervention and operational risk.
- Achieved approximately **60% reduction in manual VPN effort**, improving operational reliability, consistency, and response times.
- Performed deep-dive analysis and troubleshooting of **IKEv1/IKEv2**, **IPsec**, and VPN logs to identify and resolve complex connectivity issues.
- **Technology Stack & Tools:**
  - Python, PyCharm, IDEs | Cisco ASA (Netmiko, Paramiko) | Palo Alto (FastAPI, XML API, REST APIs, pan-device, pan-os-python) | IPsec, IKEv1/IKEv2, JSON, Logging

Technical Specialist | HCLTech — Nagpur, India | Jul 2023 - Feb 2024

- Worked on a **Palo Alto Policy Automation Platform** aimed at automating security policy management and compliance validation.
- Developed a **Python-based policy automation solution** leveraging **PAN-OS XML and REST APIs** integrated with **Panorama** for centralized firewall management.
- Integrated the automation platform with a **CI/CD pipeline using GitHub**, enabling controlled and auditable policy deployments.
- Embedded **Rapid7 vulnerability scanning** and **PCI compliance validation** directly into the automation workflow.
- Implemented **SSL VPN and IPsec site-to-site VPN monitoring**, enabling proactive issue detection and faster troubleshooting.
- Automated validation and troubleshooting of VPN connectivity and security policy behavior across environments.
- **Tools & Technologies:**
  - pan-device, pan-os-python | PAN-OS XML APIs, REST APIs | Panorama, SSL VPN, IPsec

Network - L2 Engineer | NTT Global Data Centers & Cloud Infrastructure — Mumbai, India | Aug 2022 - Jun 2023

- Key contributor to **Data Center Infrastructure Migration & Network Automation** project.
- Successfully migrated the **entire EWM data center infrastructure**, including:
  - Network devices | Servers | Storage systems
- Supported both **trading and non-trading environments**, ensuring minimal downtime and business continuity.
- Deployed and configured **Palo Alto PA-5250 firewalls** and **Cisco Nexus (N9K/N3K) switches** to enhance security, scalability, and performance.
- Implemented **CheckMK monitoring** for proactive network visibility, alerting, and operational reliability.
- Focused on building **high-performance, resilient, and secure data center networks**.

Network Security Engineer | ICEICO Technologies Pvt. Ltd. — Nagpur, India | Feb 2018 - May 2022

- Worked on **HPE Aruba Network & Security Infrastructure Implementation** projects.
- Implemented and managed enterprise-grade **network addressing and routing**, including:
  - Static routing | Dynamic routing
- Designed and configured **VLANs**, **802.1Q trunking**, **inter-VLAN routing**, and **port-channel configurations**.
- Implemented **redundancy mechanisms** such as EtherChannel, HSRP, and VPC for high availability.
- Managed and secured network traffic using **Palo Alto and Fortinet firewalls**, including:
  - Security policy configuration | Traffic monitoring | Secure traffic flow enforcement
- **Devices Managed:**
  - Palo Alto PA-5250 | Fortinet 100E | Sophos XG-115 | Aruba 2530, 2540, 2930F/M | Cisco 3810, 5400

System Administrator (Part-time | During COVID-19 Pandemic) | MRR Soft — Nagpur | Oct 2021 - Dec 2021

- Provided **internal office LAN administration** and day-to-day IT operations support.
- Supported **VPN configuration and remote access** for employees during pandemic-driven remote work.
- Performed **network troubleshooting and end-user support**, resolving connectivity and system issues.
- Handled **system updates, patching, and routine maintenance** to ensure system stability and security.